Assignment 03: External SOAP Partner Integration

Kissan Mitra • Design-time contract only • No running server or live partner gateway

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1. Context

Kissan Mitra integrates one external delivery partner through the SOAP operation **createShipment**. This partner edge is separated from the rest of Kissan Mitra's REST-oriented service APIs because the external partner contract is described by a versioned WSDL with explicit request, response, and fault messages. The SOAP Header carries partner credentials separately from the shipment request Body; production credentials must be protected in transit with TLS and managed as secrets. SOAP by itself does not guarantee message-level security or distributed transactions: any WS-Security, signing, or transaction guarantees would need to be explicitly supported and agreed with the partner.

Assignment 02 alignment

Assignment 02 defines the Delivery service as the owner of delivery records and shipment tracking. Its existing contract includes **createDelivery**, **getDeliveryStatus**, and **addTrackingEvent**. This external **createShipment** operation is a partner-facing edge used by the Delivery service; it does not replace those Kissan Mitra APIs and does not change the Orders service's ownership of order status. The Delivery service remains responsible for translating partner responses into its own delivery status and tracking data.

2. HTTP binding

The WSDL defines SOAP 1.1 over HTTP with document/literal style. The SOAPAction value below is exactly the value in the WSDL binding. The endpoint uses the reserved **.example** domain and is a design-time placeholder, not a live URL.

```
POST /soap/ShipmentService HTTP/1.1
Host: sandbox.delivery-partner.example
Content-Type: text/xml; charset=utf-8
SOAPAction: "urn:kissanmitra:delivery:CreateShipment"

[HTTP entity body: contents of soap-request.xml]
```

Endpoint URL: https://sandbox.delivery-partner.example/soap/ShipmentService

3. Discovery and catalogue entry

The Delivery service obtains the WSDL either from a partner-published, versioned contract URL or from an approved internal service catalogue. Before use, the team verifies the partner identity, endpoint ownership, TLS requirements, authentication method, contract version, and change policy. The reviewed WSDL version should be pinned with the project artefacts. No UDDI server is required or started. The tModel-style pointer below identifies the WSDL contract.

```
{
  "business": "External Delivery Partner",
  "service": "ShipmentService v1",
  "endpoint": "https://sandbox.delivery-partner.example/soap/ShipmentService",
  "wsdl": "https://sandbox.delivery-partner.example/contracts/ShipmentService-v1.wsdl",
  "tModel": {
    "name": "KissanMitra-ExternalDeliveryPartner-WSDL-v1",
    "overviewURL": "https://sandbox.delivery-partner.example/contracts/ShipmentService-v1.wsdl"
  },
  "operation": "createShipment",
  "environment": "design-time placeholder; not a live endpoint"
}
```

4. Fault mapping to the Assignment 02 contract

The partner-specific fault code **partner_unavailable** must not leak through the Kissan Mitra API. The Delivery adapter maps it to the Assignment 02 error **SERVICE_UNAVAILABLE** (HTTP 503), which is already used for unavailable downstream services in the service contracts. The Delivery service records or schedules a controlled retry using its own delivery workflow; it must not change the Orders service database or claim shipment assignment succeeded. If a retry is attempted, the merchantReference should remain stable so the partner can deduplicate repeated requests if it supports idempotency. The partner's raw fault code and XML detail should be kept out of student/client-facing API responses.

Partner outcome	Kissan Mitra-facing behavior
Successful CreateShipmentResponse	Use partnerShipmentId and trackingNumber to update the Delivery-owned delivery record/status.
partner_unavailable; retryable=true	Return/map to 503 SERVICE_UNAVAILABLE; controlled retry; do not report shipment as assigned.
Invalid request or partner validation fault	Map to the existing Delivery contract's 400 INVALID_INPUT where applicable.

5. Contract and security notes

The credential in soap-request.xml is a redacted demonstration placeholder, not a real secret. In production, use TLS, store partner credentials in a secret manager, avoid logging credentials or personal delivery data, validate response/fault namespaces, and define duplicate-request behavior for merchantReference. The SOAP messages are illustrative contract examples only. No server is implemented, no external call is made, and the example endpoint is not claimed to be reachable.

Included submission files

partner.wsdl • soap-request.xml • soap-response.xml • soap-fault.xml • integration.pdf. The WSDL includes types, messages, portType, SOAP 1.1 HTTP binding, service and port; the request, success response, and fault envelopes use the declared XML elements.